# JIANT FM · SOUND · FX · MOD

engines, effects, modulation, sequencer and songs

firmware v0.6.89 · M-VAVE FM-1 · juanjiant-bit.github.io/JIANT

## ENGINES · EDIT 1 / EDIT 2

### EIGHT ENGINES, ALL SYNTHESIS

| ENGINE | EDIT 1 | | | | EDIT 2 | | | | WHAT TO KNOW |
|---|---|---|---|---|---|---|---|---|---|
| **ANALOG** | WAVE | DETUNE | MIX | NOISE | CUT | RES | DRIVE | INT | SAW SQR TRI SIN PWM **SYNC RING SAW3**; INT: osc 2 ±24 st. Its filter is per voice. |
| **FM6** | ALG | FB | MOD | RATIO | MEG | VMOD | DETUNE | SLOT | Dexed's 6 operators, 32 algorithms drawn, .syx patches (SLOT), MOD / RATIO move the modulators. **OPERATOR · OP ENV · OPERATOR 2**: each operator on the device (ratio / fixed, fine, level, its envelope, mode, detune, velocity). **SLOT B1–B32**: a 32-voice .syx sent from the web editor. |
| **PHASE** | WAVE | WAVE2 | DCW | ENV | DETUNE | LINE | SUB | FB | Phase distortion (CZ); **FB** feeds the output back into the phase: hard edge to growl. |
| **LOFI** | CHIP | WAVE | DUTY | CRUSH | - | VIB | BEND | MASK | Chip: PLS TRI SAW NOIS WRAM. **BYTE**: ALGO VAR · MOTN GRIT BEND LOOP. **FLOAT** (floatbeat): ALGO VAR · MOTN FBK BEND LOOP; CHIP sets FULL / 4BIT / 1BIT. |
| **VOICE** | VOWEL | VOWEL2 | TALK | SHIFT | BUZZ | BREATH | Q | RAND | Formant voice (after klattsch): vowels, talk, breath. |
| **WHEEL** | REG | SUB | BODY | TOP | PERC | CLICK | DRIVE | ROTOR | Drawbar organ: registrations, percussion, rotary. |
| **NOISE** | MODE | COLOR | FREQ | PEAK | TRK | DENSE | DRIFT | CRUSH | Coloured and metallic noise, tracked or free. |
| **DRUM-X** | MORPH | TUNE | TONE | DECAY | NOISE<br>DRIVE | FM | ENV | FOLD | 8 sounds BD SD CP CH OH TM RS CB, sides A / B: **MORPH** morphs the kit. The kit first: **KIT 3** BP FEEDBK, **X-MOD** RATE DEPTH SHAPE RAND. Then a sound (KNOB 1 the sound): SIDE PITCH PMOD · DECAY NOISE COLOR · MODE PM.TYPE DRIVE · **LAYER** (no note) FEEDBK **SEND** (CHO / DLY / REV). Last LEVELS and **STEREO** IMAGE (a knob through the kit's stereo placings). |

### FILTER · EVERY ENGINE

| | | | |
|---|---|---|---|
| TYPE | CUT | RES | DRV |

**LP · HP · BP · COMB**. COMB rings tuned to the note: CUT moves it ±32 st, RES how much it rings. ANALOG and LOFI: per voice; the others: on the track, following LFO / ENV DEST FLT (DRV = the track's DIST).

### VOICE · PLAYING

| | | | |
|---|---|---|---|
| VCE | GLD | GLMOD | PRIO |

POLY · MONO · LEG · UNI; glide and its mode; LAST / LOW / HIGH priority. **8 voices** shared by the six tracks (MONO and LEGATO keep their note).

### DICE

**EDIT + SELECT**: a random factory preset of the engine, every parameter moved up to a quarter of its range. DRUM-X: both sides of every sound around the factory kit. Heard at once: OCT+ OK keeps it, OCT- CANCEL brings the sound back. SEQ / LFO / FX + SELECT: the sequence, the LFO, the effects, the same way. HOME + SELECT (after YES): a new piece: a genre, an energy (SPACE · MID · FRANTIC), its orchestration, every track a role. GLO + G4 undoes.

## MODULATION

### ENV · LFO

ENV

| | | | |
|---|---|---|---|
| ATK | DEC | SUS | REL |

ENV DEST

| | | | |
|---|---|---|---|
| FLT | PIT | SHP | LOOP |

LOOP: held, the envelope cycles from the sustain: an ADSR-shaped LFO.

LFO

| | | | |
|---|---|---|---|
| RATE | WAVE | PHS | FADE |

SIN TRI SAW SQR S&H

LFO DEST

| | | | |
|---|---|---|---|
| PIT | FLT | SHP | AMP |

### MOD MATRIX · 4 SLOTS A TRACK

| | | | |
|---|---|---|---|
| SLOT | SRC | DST | AMT |

SOURCES

LFO ENV VEL KEY RAND MODW AT EXPR M1 M2 M3 M4 STEP

DESTINATIONS

PITCH CUT SHP AMP PAN DIST CHO DLY REV RATE VIB E1...E8 CLIP PNCH

E1...E8: the engine's own EDIT knobs. CLIP / PNCH: the master.

### MSEQ · MACROS

MSEQ · A CV SEQUENCER PER TRACK

| | | | |
|---|---|---|---|
| STEP | LVL | LEN | SLEW |

16 levels, DIV on MSEQ 2, in time with the transport. Use it from MOD, SRC STEP.

MACROS M1–M4

Hold LFO: the knobs are M1–M4, latched until LFO again. Each session starts with 2 random routes a track; F3 rolls new ones, G3 clears. By hand: MOD, SRC M1...M4. Saved with the project.

### SEQUENCER

64 steps a track, up to 4 notes a step, piano roll and drum grid. **Hold a step + turn a knob**: a parameter lock. Step recording: hold a chord + KNOB 1 its length; + ARP: arpeggiated (K1 rate, K2 order, K3 oct). COMP / COMP 2: two components a step (CHANCE, RATCH, ARP NOTES / SEQ / SCALE, TRIG, RAND, OCT, TONAL, BEND, ECHO, PULSE, JUMP, REV: TIMES x1 x2 1:2, FILL, KILL); a TRIG or FILL wins; RATCH / ECHO: DIV; WHEN: each one's loops (ALWAYS 1:2 2:2 SKIP) and the step's LEN. SHIFT's CHANCE thins the whole track; C.LEN gives the components a length of their own (they drift over the steps). REC + knobs while playing: automation (64 records a section, MOTION page plays / clears them, AUTO LIST edits them).

PATTERN · SHIFT

| | | | |
|---|---|---|---|
| LEN | DIV | SWG | GATE |
| OFS | PIT | CHN | CLEN |

### SCALE · ARP

**ROOT and SCALE** are every melodic track's: CHR MAJ MIN DOR MIX PEN MPEN HARM PHRY LYD LOC MEL BLUES WHOLE DIMHW DIMWH. QNT snaps keys or the sequence, TRN transposes, **ALL TR** (SCL 2) moves every sequence along the scale. **CHORD**: a key plays a chord (TRIAD 7TH ADD9 SUS2 SUS4 in key, MAJ MIN 7 MAJ7 MIN7 POWER), VOICING LEAD leads each to the next, STRUM as a harp.

ARP MODES

OFF UP DN UPDN RND ORD REPEAT DNUP UP+8 CONV DIVG PINKY THUMB WALK CHORD **TRNS** (the keys transpose the sequence)

### SONGS

**8 projects × 4 sections (A–D)**, a song of up to 16 rows by bars (SAVE › ARRANGE; END: STOP, x2–x8, LOOP), SLOOP style. Each row can hold a **scene**: mutes, DRUM-X group mutes, macros and MIDI punch-ins. SAVE held plays sections live on the next bar, stores them, records the song (SONG REC). Each section keeps its sounds, steps, motion, kit and its SLICER.

## EFFECTS

### PER TRACK

FX · SENDS

| | | | |
|---|---|---|---|
| DST | CHO | DLY | REV |

DIST

| | | | |
|---|---|---|---|
| DST | TYPE | TONE | - |

SOFT · HARD · FOLD · CRUSH · RECT.

AUTO PAN

| | | | |
|---|---|---|---|
| DEPTH | RATE | SYNC | WAVE |

The track's pan swung L–R around its PAN. SYNC **BEAT**: RATE 4 BARS ... 1/32, in time with the bar; **FREE**: in Hz. SIN · TRI · SAW · SQR · S&H, never a click.

### SLICER · ONE ON A BUS

| | | | |
|---|---|---|---|
| SLCR | PAT | RATE | DEPTH |

OFF · GATE · STUT, 16 patterns, 1/8 ... 32T. **SL BUS**: T1–T6 and **FX** (the delay / reverb / chorus returns) ON / OFF: everything on = the whole mix chopped after the effects. **SL PITCH**: the repeats' 16-step pitch (±24 st, tape style, QNT SCL to the scale).

### DELAY · CHORUS

DLY

| | | | |
|---|---|---|---|
| TIME | FDBK | TONE | MIX |

DLY 2

| | | | |
|---|---|---|---|
| WIDE | PITCH | SPRY | MOD |

Synced or free (5 ms – 1.48 s); PITCH / SPRY: a grain delay.

CHORUS

| | | | |
|---|---|---|---|
| RATE | DEPTH | STER | - |

STER: 0 as always, down to mono, up to twice as wide.

### REVERB · FIVE MODELS

| | | | |
|---|---|---|---|
| TYPE | SIZE | DAMP | PRE |
| MOD | RATE | FILT | WIDE |

| TYPE | WHAT IT IS | SIZE | DAMP | MOD | RATE | WIDE |
|---|---|---|---|---|---|---|
| **ROOM** | Classic room | decay | darkens | chorus on the tail | its speed | stereo |
| **SPRING** | Amp spring tank | length · decay | darkens | wobble | wobble speed | stereo |
| **SHIMR** | Up an octave on every pass; **PRE → AIR** (±24 st: down darker and lower, up airier and higher) | decay | darkens | how much shimmer | — | — |
| **RESO** | 4 strings tuned to the scale; **PRE → PITCH** (0–14 steps up), **DAMP → FOLW** (OFF / T1–T6: the chord follows that track's notes) | how long they ring | FOLW | past half: the chord opens | — | — |
| **CLOUD** | Granular wash; SIZE max **freezes** it; **PRE → AIR** (±24 st: the cloud down / up, darker / airier) | decay · freeze | darkens | grains pitched + chorus | grain density | grain spread |

### MASTER

| | | | |
|---|---|---|---|
| CLIP | PNCH | DUCK | REL |

**CLIP** saturates without getting louder. **PNCH** drives the drums: denser body. **DUCK**: the kick pulls the rest down (REL its release). An always-on **leveler** (slow 2:1, –9...+6 dB) before the limiter keeps patches at a steady volume.

PUNCH-IN FX (FX HELD)

REPEAT 1/8 · 1/16 · 1/32, LPF, HPF; on the notes: OCT–/+, ½ TEMPO, DEC–/+, STUTTER, ATK+, ARP, RANDOM. Knobs: FILTER CRUSH THROW DEPTH. Recordable per section.

GPL-3.0 · fork of Felucca (Hügelton Instruments) · song mode from SLOOP